INTERNATIONAL DEVELOPMENT ASSOCIATION

CAPE VERDE

Joint Fund-Bank Debt Sustainability Analysis

Prepared by the staffs of the International Development Association and the International Monetary Fund

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The risk of debt distress in Cape Verde remains low*. Domestic debt is expected to continue to decline owing to a prudent fiscal stance in line with the authorities' commitment with the Fund and dialogue with World Bank. Moreover, all external debt indicators remain below the relevant thresholds under the baseline and alternative scenarios, even after concessional external financing declines as the country graduates from the United Nations' least-developed-country (LDC) status in 2008. Continued flow of foreign direct investment is projected to finance a large part of the external current account deficit.*

IV. BACKGROUND[13]

31. **This DSA updates the DSA done in 2006 at the inception of the Policy Support Instrument (PSI) of the IMF.**[14] This is the second DSA for Cape Verde that applies the methodology of Fund-World Bank debt sustainability framework (DSF) for low income countries.[15] In accordance with the framework, this DSA compares the expected trajectory of debt indicators with the thresholds for countries with strong policies and institutions (Table 4).[16] Based on 2006 debt estimates updated during the mission for the third review of the PSI in October 2007, the analysis maintains the previous assessment that the risk of debt distress in Cape Verde remains low.

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[13] This analysis includes only central government debt and guarantees, thus excluding municipalities and state-owned enterprises.

[14] Appendix II of IMF Country Report No. 06/334, 09/2005.

[15] Although Cape Verde is not a low-income country and is not eligible for HIPC assistance and MDRI, it is eligible for PRGF and IDA financing based on the small island exception. In July 2007 Cape Verde became eligible to (limited) IBRD lending in addition to IDA, thus becoming a blend country. After graduation from LDC status in 2008, it will continue to be a blend country.

[16] Cape Verde's average score under the World Bank's Country Policy and Institutional Assessment (CPIA) in 2004-6 is 4.1, above the floor for strong performers of 3.75.

32. **In 2006, the total central government net debt and guarantees were reduced markedly as a percentage of GDP.** Although the guarantees increased, the total nominal debt fell to 77 percent of GDP in 2006, down from 87 percent in 2005. Rapid GDP growth in 2006 and prudent fiscal policy led to this reduction (Table 8). The central government external debt was reduced from 54 percent of GDP to 48 percent during 2006 while its domestic debt was reduced from 33 percent of GDP to 29 percent.

33. **Multilateral and bilateral agencies continue to be the largest external creditors.** Commercial creditors account for less than 2 percent of the external debt. The World Bank (IDA) was the largest multilateral creditor, followed by the African Development Fund. Portugal is the main bilateral lender (Table 1).

Table 1. Cape Verde: Central government external debt, 2004-06

	2004	2005	2006
	Actual		Est.
	(In percent of the nominal debt)		
Total	100	100	100
Multilaterals	77	78	78
IMF	2	2	2
IDA	40	42	44
Others	35	33	32
Official Bilaterals	19	19	19
Paris Club members	12	13	12
Non-Paris Club members	4	5	7
Commercial	4	3	2
Memorandum: Total nominal external debt of the central government			
In US$ million [1]	553	513	598
In percent of GDP	55	54	48

Source: Cape Verdian authorities, staff estimates and projections.

V. MEDIUM-TERM MACROECONOMIC FRAMEWORK AND ASSUMPTIONS OF THE DSA

34. **In the baseline macroeconomic framework of this revised DSA, economic growth is projected to be slightly higher than assumed in the previous DSA (Table 2).** Real GDP growth is expected to average about 7 percent through 2012 underpinned by continued robust growth in the services and construction sectors driven by tourism. Tourism arrivals have increased and several large projects in tourism entered the pipeline since the last DSA. The government's strategy involves promoting tourism by expanding infrastructure, improving the business environment. As result, the investment promotion agency has signed investment protocols with several private hotel operators. This gives reasonable confidence that many of these investments will indeed take place. As the initial phase of the tourism boom tapers off, GDP growth is conservatively assumed to slow gradually to 5 percent over the long term, below historical levels. However, the authorities are of the view that the growth rates driven by the tourism boom will be stronger and last longer than this DSA's assumptions.

Consistent with the peg to the Euro, annual inflation is expected to stay between 2 and 3 percent in line with that of Cape Verde's main trading partners.

35. **Strong FDI inflows are assumed to finance the higher current account deficit and limit the need for new external borrowing.** The increase in the current account deficit reflects the growth of imports for capital formation and tourism (Table 2). Additionally, during the execution of the Millennium Challenge Corporation (MCC) compact, imports will be temporarily above the trend. FDI is assumed to remain above 10 percent of GDP until 2015 and to gradually decline after the investment boom fades. Tourism exports will continue to rise faster than GDP throughout the projection period, evidencing the specialization of Cape Verde in tourism. Tourism growth will compensate the increase in imports and the decline in remittances. It is assumed that private current transfers, including emigrant remittances, will decline from 20 percent of GDP in 2006 to 12 percent in 2027 as incomes in Cape Verde grow faster than in emigrant recipient countries and emigration recedes.

36. **Prudent fiscal policy is expected to continue in line with the PSI objectives, which will help preserve the low risk of debt distress.** In the baseline scenario, revenue growth and expenditure restraint will maintain the primary fiscal deficit below the debt-stabilizing deficit throughout the projection period (Table 8). Tax revenues are expected to be buoyant underpinned by both robust economic growth and improvements in tax administration, as well as rationalization of tax exemptions and broadening of the tax base, which will compensate for the expected decline in grants. Improvements in budget execution and monitoring, including under the IMF's PSI, are assumed to induce further restraint on current expenditures. Capital expenditures are assumed to remain above 10 percent of GDP throughout the projection period, in line with the authorities' intentions to improve the country's infrastructure (Table 3).

Table 2. Cape Verde: Macroeconomic Baseline Assumptions, 2007-27
(in percent, unless otherwise indicated)

	Average (1996-2006)	5 years ahead		6-20 years ahead	
		Previous DSA	Current DSA	Previous DSA	Current DSA
Real GDP growth	6.8	6.2	7.3	5.1	5.2
Inflation	2.6	2.2	2.3	2.1	2.0
Exports of G&S growth (in US dollars)	14.8	6.9	15.2	8.5	9.7
Imports of G&S growth (in US dollars)	8.9	9.0	14.0	6.0	8.4
Private transfers, incl. remittances (percent of GDP)	19.4	20.5	17.3	11.5	13.7
Current account balance (percent of GDP)	-9.3	-9.7	-13.2	-7.5	-12.0
Foreign direct investment (percent of GDP)	3.5	3.2	12.2	2.0	9.3
Current acct. and FDI (Percent of GDP)	-5.8	-6.5	-1.0	-5.5	-2.6
Grant element of new external borrowing	…	28.3	27.9	9.3	9.5
Exchange rate, nominal currency per US dollar, p.a	101.3	92.7	79.4	92.1	78.7
Public sector revenue and grants (percent of GDP)	30.3	34.1	29.2	28.8	28.0

Source: Cape Verdian authorities, staff estimates and projections.

37. **Grants and external financing assumptions under this revised DSA are conservative, in line with Cape Verde's impending graduation from LDC status.**

Assumptions on concessional financing and grants have been significantly revised downwards compared to the previous DSA. It is now assumed that as Cape Verde graduates from LDC status in 2008, grants as a percentage of GDP and the degree of concessionality of the new borrowing will decline (Table 6 and 8, Figure 1). The share of commercial financing in total new external borrowing is assumed to gradually increase to 85 percent until 2027, and the grant element of new external borrowing will decline to zero. Because commercial loans have shorter maturity, the debt service will also increase (Table 7, Figure 1, bottom charts). The lower concessionality along with an assumed decrease in current transfers will contribute to the widening of the current account deficit. The interest rate on the domestic debt remains constant at 4.3 percent throughout the period, and the marginal borrowing in the stress scenarios is assumed to be on commercial terms (Table 7).

Table 3. Cape Verde: Comparison of Scenarios, 2008-12

	Scenario	
	Baseline	Scaling-up
	(Percent of GDP, period average)	
Real GDP growth, 2007-2027	5.8	5.8
Imports of G&S (in US$)	78.8	72.4
Capital expenditures	12.6	14.5
Fiscal balance	-3.2	-5.2
External current account balance	-12.1	-12.3

VI. External Central Government Debt Sustainability

A. Baseline scenario

38. **Cape Verde's disciplined macroeconomic policy framework, continued strong FDI flows, and the resulting low reliance on external central government debt will maintain low risk of debt distress.** The ratio of NPV of the external debt-to-GDP is expected to decline during most of the period because the external borrowing-to-GDP ratio generally stays below the debt-stabilizing ratio (Tables 4, 6 and 7, Figure 1). The external borrowing is small because FDI finances a large part of the higher current account deficit (Tables 2 and 6). Therefore, the NPV of the central government external debt declines from 26 percent of GDP in 2007 to below 22 percent of GDP after 2009, comfortably below the debt-burden threshold. The ratios of debt-to-exports and to revenues also remain below the indicative thresholds. Because commercial loans have shorter maturities, debt service ratios increase. Nevertheless, the debt service indicators remain below the thresholds, suggesting that roll-over difficulties are unlikely (Table 7 and Figure 1). The assessment of low risk hinges on the assumption that tourism exports and FDI flows are sufficient to finance the imports used as inputs in the tourism industry. This assumption is reasonable because tourism exports are partly produced with domestic factors.

B. Alternative scenarios and stress tests

39. **The government's external debt indicators remain well below the thresholds in all scenarios.** The most extreme stress scenario is a hypothetical 30-percent devaluation of the exchange rate in 2008. It causes a one-off increase of 7 percent of GDP in the NPV of the debt. However, even in this stress scenario the external debt ratios remain below the threshold as the NPV of the external debt before the shock is low.

Table 4. Cape Verde: Central Government External Debt Ratios

		Baseline			Scale-up scenario		
	Thresholds [1]	2007	2017	2027	2007	2017	2027
NPV of debt in percent of:							
GDP	50	26	21	18	23	23	18
Exports	200	60	37	28	60	44	31
Revenues [2]	300	84	75	63	84	89	70
Debt service in percent of:							
Exports	25	4	3	6	4	4	6
Revenues [2]	35	5	7	12	5	8	13

Source: Ministry of Finance; and staff extimates.

[1] Based on Cape Verde's 2004-06 classification as a strong performer.

[2] Including grants.

C. Scaling-up scenario

40. **An alternative scenario of scaling-up borrowing under commercial terms and capital expenditures has been simulated (Table 3).** Although the authorities have not yet finalized their medium-term investment program, they intend to explore new mechanisms to scale-up infrastructure investment. To show the robustness of the debt sustainability under the scaling-up scenario, this alternative scenario assumes that Cape Verde will borrow US$ 35 million on commercial terms in each of the five years covering 2008-12 in addition to the borrowing in the baseline scenario.[17] This extra borrowing is used to scale-up public capital expenditure and is matched by a corresponding increase in imports. This scenario simulates, for example, additional commercial borrowing to invest in the electricity sector in 2008 and the envisaged borrowing from the IBRD. The additional interest expense arising from the scaling up of external borrowing is financed with domestic borrowing. For prudence, economic growth is maintained at the same level of the baseline scenario.

41. **The risk of debt distress also remains low in the scaling-up scenario**. All ratios remain well below the thresholds throughout the forecast horizon. The additional

[17] This value corresponds to the ceiling for non-concessional external borrowing agreed under the PSI.

US$35 million borrowed in this scenario, equivalent to 2 percent of GDP on average during 2008-12, does not represent an important burden given that the debt is low and that GDP growth rate is still higher than the debt accumulation rate. Further, the debt ratios remain below threshold even if shocks are simulated to hit on top of the scaling up scenario.

VII. TOTAL CENTRAL GOVERNMENT DEBT SUSTAINABILITY

A. Baseline scenario

42. **In line with the PSI objectives, the total central government debt and guarantees remain below current level throughout the forecast horizon in the baseline scenario.** This reflects the Bank and Fund staff's expectation that the good fiscal policy performance will be maintained in the long term. In addition to the expected decline in the external debt ratio discussed previously, the domestic central government debt ratio is reduced to below 20 percent of GDP by 2008, one year ahead of the PSI target (Table 8). The low fiscal deficits and the contribution of the GDP growth to the debt ratio are the main reason for the decline. After 2013, the total central government debt-to-GDP ratio stabilizes around 34 percent of GDP (Figure 2). The ratios of debt-to-revenue and debt service-to-revenue also suggest that roll-over difficulties are unlikely to arise.

B. Alternative scenarios and stress tests

43. **In the long term, endogenous risks to the sustainability of the total central government debt are more threatening than exogenous shocks.** The current policies are consistent with maintaining a low risk of debt distress, which is evidenced by the fact that the debt ratios remain at low levels in the scenario of unchanged balance from the 2007 level (Table 9, and Figure 2). However, there is the endogenous risk that prudent fiscal policies will not be maintained in the long term. In this scenario, the primary deficit returns to the historical levels, which is substantially higher than the debt-stabilizing level. Therefore, in this scenario the debt rises sharply (scenario A1 of Table 9, Figure 2).[18]

44. **An exogenous shock alone does not increase the risk of debt distress to a significant level.** The most extreme test is one in which an exogenous shock temporarily widens the primary deficit to 11 percent of GDP[19] during 2008-09 (scenario B2 in Table 9,

[18] This historical scenario assumes also that GDP growth is at historical average of 7 percent, more optimistic than baseline projections.

[19] Equivalent to historical balance minus one standard deviation.

Table 5. Total Central Government Debt Ratios, 2007-27

	Baseline			Scale-up scenario		
	2007	2017	2027	2007	2017	2027
NPV of total debt in percent of:						
GDP	47	33	34	47	41	40
o/w domestic debt	23	14	17	23	17	21
Revenues [1]	152	119	120	152	146	143
Debt service in percent of:						
Revenues [1]	11	11	18	11	14	20

Source:Ministry of Finance; and staff estimates.

[1] Including grants.

Figure 2). Because in this scenario the debt ratios increase only temporarily, the low risk of debt distress is preserved throughout the period. This conclusion applies also to the situation where the extreme shock hits on top of the scale up scenario.

C. Scaling-up scenario

45. **Under the scaling up of borrowing and public investment, the low risk of debt distress is preserved**. The fiscal deficit during the 5 years of the scaling up is smaller than the debt-stabilizing deficit, causing the domestic debt to remain around the PSI target of 20 percent of GDP during the projection period (Table 5). All the other total central government debt indicators remain within reasonable levels in the scaling up scenario, even if shocks are simulated to hit on top of the scaling up scenario (Tables 5 and 8, and Figure 2).

VIII. CONCLUSION

46. **The debt sustainability analysis concludes that the risk of debt distress in Cape Verde is low.** This is notwithstanding the conservative assumptions made in the baseline scenario including on financing terms. The conclusion hinges on the assumption that FDI flows and tourism exports will fully offset the growth of imports caused by them. The most important threat to debt distress risk is a return to historical fiscal deficits, which highlights the need for continued fiscal prudence. Based on the expectation that the fiscal prudence will continue during the projection period, the assessment of distress risk is low. The risk of debt distress for Cape Verde remains low also under large exogenous shocks and under a 5-year scaling up of capital investment.

Table 6. Cape Verde: External Debt Sustainability, Baseline Scenario, 2004-2027 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2004	2005	2006			2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External debt (nominal) 1/	**57.4**	**58.7**	**52.8**			**49.4**	**44.9**	**41.6**	**38.9**	**36.9**	**35.7**		**30.9**	**21.3**	
o/w central government	54.6	53.8	48.0			44.2	40.2	37.4	34.9	33.3	32.4		28.5	19.9	
Change in external debt	-1.6	1.4	-5.9			-3.5	-4.4	-3.3	-2.8	-2.0	-1.2		-0.9	-1.2	
Identified net debt-creating flows	-0.1	-8.8	-13.1			-2.4	-2.8	-2.2	-0.7	-1.8	-1.3		0.5	3.8	
Non-interest current account deficit	**13.7**	**2.8**	**4.6**	**9.5**	**3.6**	**9.0**	**11.0**	**12.1**	**14.2**	**13.1**	**13.4**	**12.1**	**12.4**	**9.7**	**11.4**
Deficit in balance of goods and services	37.6	27.8	25.8			29.1	29.3	31.0	32.4	30.2	29.9		25.9	20.3	
Exports	32.0	36.5	42.3			43.3	45.6	47.2	48.2	49.4	50.8		56.0	62.4	
Imports	69.6	64.3	68.1			72.4	74.9	78.2	80.6	79.7	80.7		81.9	82.7	
Net current transfers (negative = inflow)	-25.2	-27.8	-24.6	-25.8	1.4	-22.3	-20.6	-21.3	-20.8	-20.1	-19.6		-17.2	-13.9	-16.2
o/w official	-5.7	-4.6	-4.1			-4.8	-2.7	-3.6	-3.5	-3.2	-3.1		-2.6	-2.3	
Other current account flows (negative = net inflow)	1.3	2.7	3.3			2.3	2.3	2.4	2.6	3.0	3.1		3.7	3.2	
Net FDI (negative = inflow)	**-7.4**	**-7.5**	**-9.4**	**-5.9**	**3.1**	**-8.8**	**-11.0**	**-11.8**	**-12.6**	**-12.9**	**-12.9**	**-11.7**	**-11.0**	**-5.5**	**-9.3**
Endogenous debt dynamics 2/	**-6.4**	**-4.0**	**-8.3**			**-2.6**	**-2.9**	**-2.5**	**-2.3**	**-2.0**	**-1.8**		**-0.9**	**-0.4**	
Contribution from nominal interest rate	0.7	0.6	0.5			0.5	0.5	0.5	0.5	0.5	0.5		0.6	0.7	
Contribution from real GDP growth	-2.2	-3.4	-5.4			-3.1	-3.3	-3.0	-2.7	-2.4	-2.3	**-2.8**	-1.5	-1.0	**-1.2**
Contribution from price and exchange rate changes	-4.8	-1.2	-3.4			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	-1.5	10.2	7.2			-1.1	-1.6	-1.1	-2.1	-0.2	0.1		-1.4	-5.0	
o/w exceptional financing	-1.2	3.6	-6.6			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	**...**	**...**	**34.0**			**31.0**	**28.5**	**26.7**	**25.3**	**24.5**	**24.3**		**23.4**	**18.9**	
In percent of exports	...	...	80.2			71.6	62.4	56.5	52.5	49.6	47.8		41.7	30.3	
NPV of central gov. external debt and guarantees	**...**	**...**	**29.1**			**25.8**	**23.8**	**22.4**	**21.4**	**20.9**	**21.0**		**21.0**	**17.5**	
In percent of exports	...	...	68.9			59.6	52.1	47.4	44.3	42.3	41.3		37.5	28.1	
In percent of government revenues	...	...	97.5			83.8	82.7	74.6	70.2	74.0	74.2		74.9	62.6	
Debt service-to-exports ratio (in percent)	11.2	8.6	5.7			5.0	4.5	4.3	3.8	3.5	3.3		4.3	6.6	
PPG debt service-to-exports ratio (in percent)	11.2	8.6	5.7			5.0	4.5	4.3	3.8	3.5	3.3		4.3	6.6	
PPG debt service-to-revenue ratio (in percent)	11.2	10.4	8.0			7.0	7.1	6.7	6.0	6.2	5.9		8.6	14.6	
Total gross financing need (billions of U.S. dollars)	0.1	0.0	0.0			0.0	0.0	0.0	0.1	0.0	0.1		0.1	0.7	
Non-interest current account deficit that stabilizes debt ratio	**15.3**	**1.4**	**10.5**			**12.5**	**15.5**	**15.4**	**17.0**	**15.1**	**14.6**	**15.0**	**13.3**	**10.9**	**12.3**
Key macroeconomic assumptions															
Real GDP growth (in percent)	4.3	6.5	10.8	7.3	2.5	6.9	7.7	7.4	7.4	7.1	6.8	7.2	5.0	5.0	5.2
GDP deflator in US dollar terms (change in percent)	8.9	2.2	6.1	2.0	11.0	11.8	6.6	4.3	4.2	4.9	3.7	5.9	3.0	2.8	2.9
Effective interest rate (percent) 5/	1.3	1.2	1.0	12.4	14.5	1.0	1.1	1.2	1.3	1.3	1.4	1.2	1.9	3.1	2.2
Growth of exports of G&S (US dollar terms, in percent)	15.9	24.1	36.3	18.1	13.9	22.4	20.9	16.0	14.2	15.2	13.8	17.1	10.3	8.1	9.7
Growth of imports of G&S (US dollar terms, in percent)	17.2	0.6	24.6	11.8	11.6	27.0	18.9	17.0	15.4	11.0	12.2	16.9	8.3	8.0	8.4
Grant element of new public sector borrowing (in percent)	**...**	**...**	**...**	**...**	**...**	**37.5**	**35.7**	**30.9**	**27.1**	**24.1**	**21.6**	**29.5**	**13.1**	**0.7**	**9.5**
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.4	1.2	0.9	0.8	0.8	0.8		0.4	0.0	0.3
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			37.6	35.8	31.0	27.3	24.2	21.7		13.1	0.8	9.5
Memorandum items:															
Nominal GDP (billions of US dollars)	0.9	1.0	1.2			1.4	1.6	1.8	2.0	2.3	2.5		3.9	8.3	
(NPVt-NPVt-1)/GDPt-1 (in percent)						1.7	1.5	1.3	1.5	2.1	2.3	1.7	1.6	0.6	1.4

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Table 7. Cape Verde: Sensitivity Analyses for Key Indicators of Central Government External Debt and Guarantees, 2007-27
(In percent)

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
NPV of debt-to-GDP ratio								
Baseline	26	24	22	21	21	21	**21**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	26	28	29	28	30	30	**28**	16
A2. New public sector loans on less favorable terms in 2008-27 2/	26	25	24	23	23	24	**26**	26
Scaling-up scenario	23	24	24	24	25	25	**23**	18
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	26	24	24	23	22	22	**22**	18
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	26	30	38	32	28	24	**21**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	26	28	30	29	28	28	**28**	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09	26	28	32	28	25	23	**21**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	26	29	37	33	30	28	**26**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	26	33	31	30	29	29	**29**	24
NPV of debt-to-exports ratio								
Baseline	60	52	47	44	42	41	**37**	28
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	60	61	62	59	60	59	**50**	25
A2. New public sector loans on less favorable terms in 2007-26 2/	60	54	50	48	47	47	**47**	42
Scaling-up scenario	60	58	57	56	56	55	**44**	31
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	60	52	47	44	42	41	**37**	28
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	60	77	103	86	72	61	**48**	36
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	60	52	47	44	42	41	**37**	28
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09	60	62	67	58	50	45	**37**	28
B5. Combination of B1-B4 using one-half standard deviation shocks	60	61	73	64	56	51	**43**	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	60	52	47	44	42	41	**37**	28
NPV of debt-to-revenue ratio								
Baseline	84	83	75	70	74	74	**75**	63
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	84	98	97	94	105	106	**99**	56
A2. New public sector loans on less favorable terms in 2007-26 2/	84	85	79	77	83	85	**94**	94
Scaling-up scenario	84	92	89	89	97	98	**89**	70
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	84	85	79	74	78	78	**79**	66
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	84	105	126	106	98	85	**75**	63
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	84	97	100	94	99	99	**100**	84
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09	84	98	105	91	88	80	**75**	63
B5. Combination of B1-B4 using one-half standard deviation shocks	84	100	124	109	106	99	**92**	77
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	84	115	104	98	103	103	**104**	87

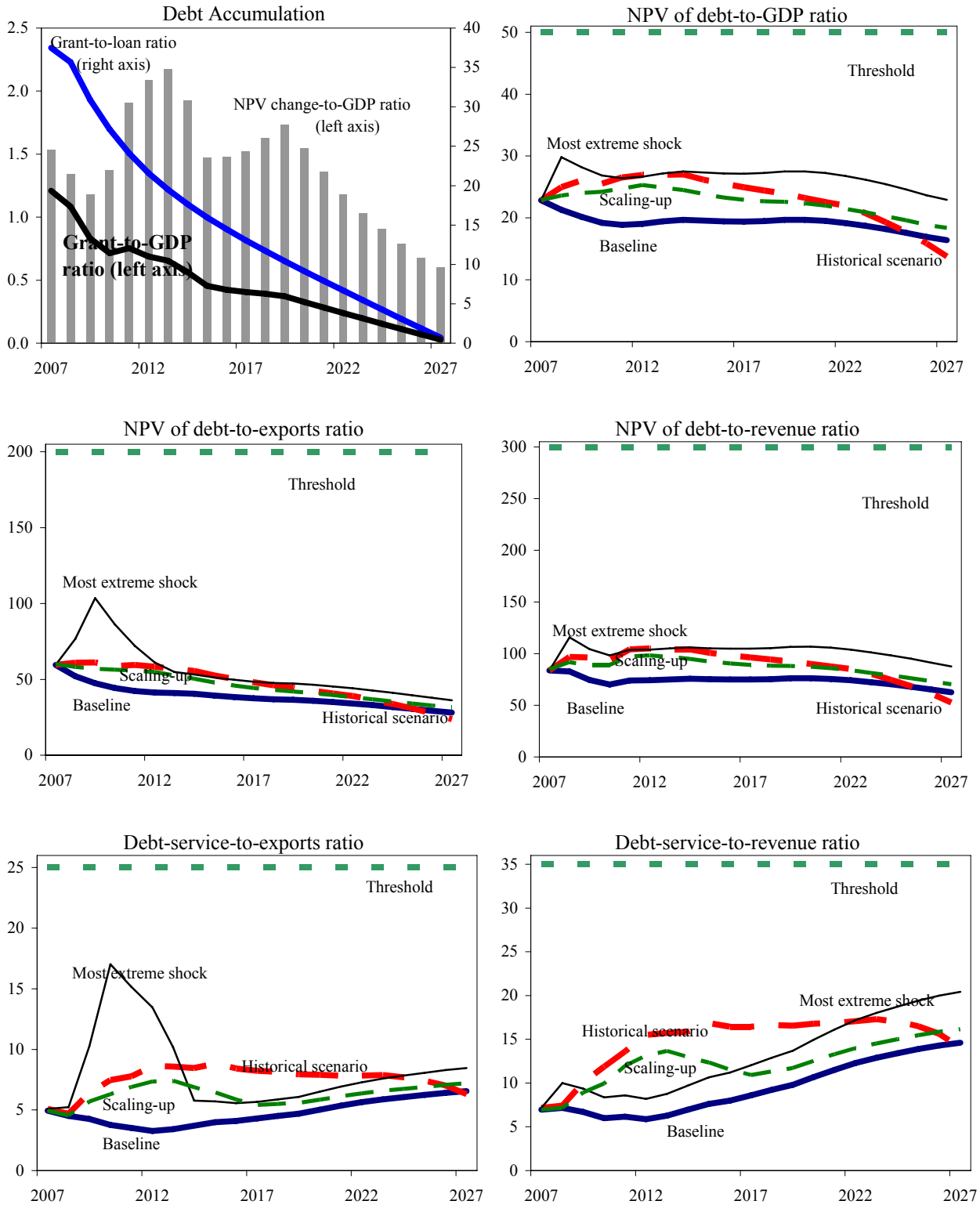
Table 7. Cape Verde: Sensitivity Analyses for Key Indicators of Central Government External Debt and Guarantees, 2007-27 (continued)
(In percent)

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
Debt service-to-exports ratio								
Baseline	5	5	4	4	4	3	**4**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	5	5	6	7	8	9	**8**	7
A2. New public sector loans on less favorable terms in 2008-27 2/	5	5	4	4	3	3	**4**	11
Scaling-up scenario	5	5	6	6	7	7	**5**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	5	5	4	4	4	3	**4**	7
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	5	5	10	17	15	13	**6**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	5	5	4	4	4	3	**4**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09	5	5	7	9	8	8	**4**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	6	10	9	8	**5**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	5	5	4	4	4	3	**4**	7
Debt service-to-revenue ratio								
Baseline	7	7	7	6	6	6	**9**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	7	7	10	12	13	15	**17**	15
A2. New public sector loans on less favorable terms in 2008-27 2/	7	7	6	6	6	5	**9**	24
Scaling-up scenario	7	7	9	10	12	13	**11**	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	7	7	7	6	6	6	**9**	15
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	7	7	12	21	21	19	**9**	15
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	7	8	9	8	8	8	**12**	20
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09	7	7	11	15	15	14	**9**	15
B5. Combination of B1-B4 using one-half standard deviation shocks	7	8	10	17	16	15	**11**	18
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	7	10	9	8	9	8	**12**	20
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	-2	-2	-2	-2	-2	-2	**-2**	-2

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Cape Verde: Indicators of Central Government External Debt and Guarantees Under Alternative Scenarios, 2007-2027



Source: Staff projections and simulations.

Table 8. Cape Verde: Central Government Debt Sustainability, Baseline Scenario, 2004-2027
(In percent of GDP, unless otherwise indicated)

	Actual					Projections									
	2004	2005	2006	Historical Average 5/	Standard Deviation 5/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Central government debt 1/	**88.9**	**87.1**	**77.3**			**66.4**	**59.6**	**54.3**	**50.1**	**47.4**	**45.8**		**42.5**	**36.6**	
o/w foreign-currency denominated	54.6	53.8	48.0			43.5	40.2	37.2	34.8	33.1	32.4		28.5	19.9	
o/w domestic debt	34.3	33.3	29.3			22.9	19.4	17.1	15.3	14.3	13.5		14.0	16.7	
Change in debt	3.6	-1.8	-9.8			-10.9	-6.8	-5.3	-4.2	-2.7	-1.6		-0.4	-1.0	
Identified debt-creating flows	-2.9	7.1	-12.3			-6.4	-4.2	-3.3	-3.3	-2.0	-1.0		-0.2	-1.0	
Primary deficit	1.3	4.2	3.2	5.8	5.3	1.3	1.9	1.5	1.0	2.1	2.1	**1.6**	1.7	0.2	**1.3**
Revenue and grants	32.1	30.0	29.9			30.8	28.7	30.0	30.4	28.3	28.3		28.0	28.0	
of which : grants	9.0	6.3	5.6	7.8		6.7	4.9	6.0	6.1	4.0	3.9	**5.3**	3.4	3.1	**3.3**
Primary (noninterest) expenditure	33.4	34.2	33.1			32.1	30.7	31.5	31.4	30.3	30.4		29.7	28.2	
Automatic debt dynamics	-4.2	2.9	-15.5			-7.8	-6.1	-4.8	-4.3	-4.0	-3.2		-1.9	-1.2	
Contribution from interest rate/growth differential	-2.1	-5.3	-9.4			-5.3	-4.9	-3.9	-3.5	-3.1	-2.8		-1.6	-1.1	
of which : contribution from average real interest rate	1.4	0.1	-1.0			-0.3	-0.1	0.2	0.2	0.2	0.2		0.4	0.7	
of which : contribution from real GDP growth	-3.5	-5.4	-8.5			-5.0	-4.8	-4.1	-3.7	-3.3	-3.0		-2.0	-1.8	
Contribution from real exchange rate depreciation	-2.0	8.2	-6.0			-2.4	-1.2	-0.8	-0.8	-0.9	-0.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	6.5	-8.9	2.5			-4.5	-2.6	-1.9	-0.9	-0.7	-0.5		-0.2	0.0	
NPV of central government debt	**34.3**	**33.3**	**60.5**			**46.9**	**41.8**	**38.2**	**35.2**	**33.2**	**32.4**		**33.4**	**33.7**	**34.0**
o/w foreign-currency denominated	0.0	0.0	31.2			24.0	22.4	21.2	19.9	18.9	18.9		19.4	16.9	
o/w external	...	...	31.2			24.0	22.4	21.2	19.9	18.9	18.9		19.4	16.9	
NPV of contingent liabilities (not included in debt)	1.7	1.6	5.7			5.1	4.6	4.1	3.7	3.3	3.0		2.0	0.9	
Gross financing need 2/	37.0	37.9	34.8			29.7	27.5	24.8	22.4	21.3	20.1		19.0	21.2	
NPV of central government debt-to-revenue and grants ratio (in percent)	107.0	111.0	202.4			152	146	127	116	118	114		119	120	
NPV of central government debt-to-revenue ratio (in percent)	148.4	140.7	248.9			195	176	159	145	137	133		136	135	
o/w external 3/	…	…	128.5			100	94	88	82	78	78		79	68	
Debt service-to-revenue and grants ratio (in percent) 4/	21.1	19.7	15.8			10.7	11.1	10.3	9.2	9.3	8.8		11.4	17.7	
Debt service-to-revenue ratio (in percent) 4/	29.3	24.9	19.5			13.7	13.4	12.8	11.5	10.8	10.2		13.0	20.0	
Primary deficit that stabilizes the debt-to-GDP ratio	-2.3	6.0	13.0			12.2	8.7	6.7	5.2	4.8	3.7	**6.9**	2.1	1.2	**1.9**
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	4.3	6.5	10.8	7.3	2.5	6.9	7.7	7.4	7.4	7.1	6.8	**7.2**	5.0	5.0	**5.2**
Average nominal interest rate on forex debt (in percent)	1.2	1.2	1.1	1.3	0.3	1.2	1.2	1.3	1.4	1.5	1.6	**1.4**	2.0	3.3	**2.4**
Average real interest rate on domestic currency debt (in percent)	7.6	2.8	-0.4	4.2	4.3	0.2	0.8	2.3	2.7	2.4	2.6	**1.8**	3.1	2.8	**3.1**
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.8	16.4	-12.6	0.5	12.9	-5.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-1.0	2.0	5.1	2.2	3.0	4.0	4.8	3.8	3.5	4.0	3.7	**4.0**	3.0	2.8	**2.9**
Growth of real primary spending (deflated by GDP deflator, in percent)	18.8	9.2	7.1	5.8	18.1	3.9	2.9	10.3	7.0	3.4	6.9	**5.7**	5.0	4.6	**4.6**
Grant element of new external borrowing (in percent)	...	...	...	…	…	18.2	35.7	30.9	27.1	24.1	21.6	**26.3**	13.1	0.7	**9.5**

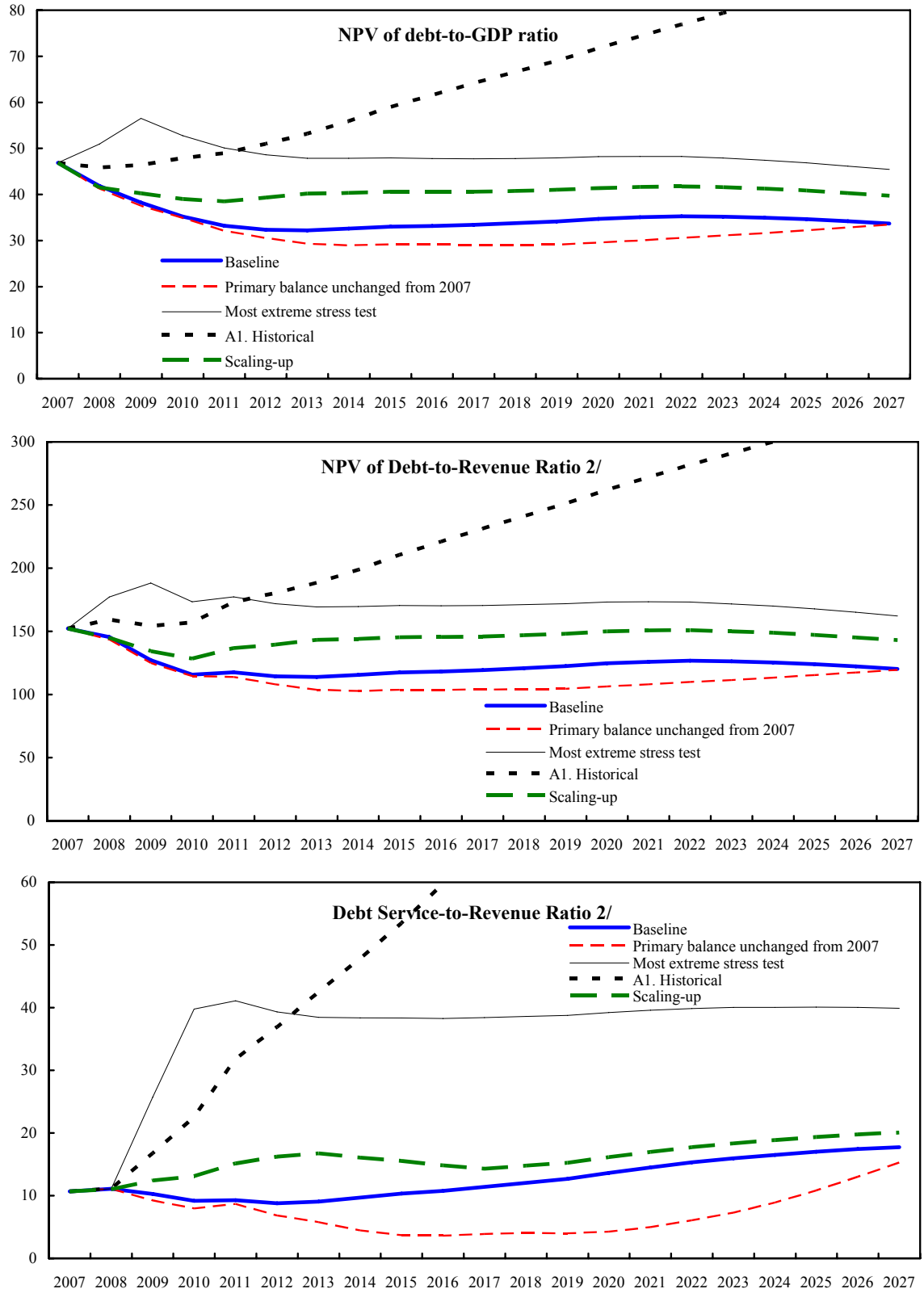
Sources: Country authorities; and Fund staff estimates and projections.
1/ Includes only central government. Gross debt is used.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 9. Cape Verde: Sensitivity Analysis for Key Indicators of Central Government Debt, 2007-2027

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
NPV of Debt-to-GDP Ratio								
Baseline	47	42	38	35	33	32	**33**	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	46	46	48	49	51	**64**	89
A2. Primary balance is unchanged from 2007	47	41	38	35	32	31	**29**	33
A3. Permanently lower GDP growth 1/	47	42	39	37	35	35	**42**	63
Scaling-up scenario	47	42	40	39	39	39	**41**	40
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviatio	47	44	42	40	39	40	**46**	57
B2. Primary balance is at historical average minus one standard deviation:	47	51	57	53	50	49	**48**	45
B3. Combination of B1-B2 using one half standard deviation shocks	47	49	53	49	46	44	**43**	41
B4. One-time 30 percent real depreciation in 2008	47	51	46	43	40	38	**38**	40
B5. 10 percent of GDP increase in other debt-creating flows in 2008	47	52	48	44	42	41	**41**	40
NPV of Debt-to-Revenue Ratio 2/								
Baseline	152	146	127	116	118	114	**119**	120
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	152	159	154	157	173	180	**231**	326
A2. Primary balance is unchanged from 2007	152	143	125	115	114	108	**104**	120
A3. Permanently lower GDP growth 1/	152	147	130	120	124	124	**149**	224
Scaling-up scenario	152	145	134	129	137	140	**146**	143
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviatio	152	151	139	131	138	139	**165**	202
B2. Primary balance is at historical average minus one standard deviation:	152	177	188	173	177	172	**170**	162
B3. Combination of B1-B2 using one half standard deviation shocks	152	170	174	159	162	157	**155**	146
B4. One-time 30 percent real depreciation in 2008	152	177	155	140	140	135	**137**	141
B5. 10 percent of GDP increase in other debt-creating flows in 2008	152	180	159	146	149	145	**146**	142
Debt Service-to-Revenue Ratio 2/								
Baseline	11	11	10	9	9	9	**11**	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	11	11	17	23	32	37	**66**	126
A2. Primary balance is unchanged from 2007	11	11	9	8	9	7	**4**	15
A3. Permanently lower GDP growth 1/	11	11	11	10	11	11	**22**	63
Scaling-up scenario	11	11	12	13	15	16	**14**	20
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviatio	11	11	12	13	15	17	**30**	56
B2. Primary balance is at historical average minus one standard deviation:	11	11	26	40	41	39	**38**	40
B3. Combination of B1-B2 using one half standard deviation shocks	11	11	22	31	32	30	**29**	30
B4. One-time 30 percent real depreciation in 2008	11	11	11	10	11	11	**15**	26
B5. 10 percent of GDP increase in other debt-creating flows in 2008	11	11	27	25	26	25	**26**	29

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projectio
2/ Revenues are defined inclusive of grants.

Figure 2. Cape Verde: Indicators of Public Debt Under Alternative Scenarios, 2007-2027 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2017.
2/ Revenue including grants.